M. Sc. Physics (Semester-II) 2026
Quantum Mechanics II – Tutorial 10

Tutorial exercises

1. Consider scattering by a repulsive δ-shell potential: $V(r) = \gamma \frac{\hbar^2}{2m}\delta(r-R)$, $\gamma > 0$. Set up an equation that determines the s-wave phase shift δ_0 as a function of $k(E = \hbar^2 k^2/2m)$.

2. Consider the scattering of a particle by an impenetrable sphere

$$V(r) = 0 \qquad r > a \tag{1}$$
$$= \infty \qquad r < a \tag{2}$$

   - Derive an expression for the s-wave ($\ell = 0$) phase shift. (You need not know the detailed properties of the spherical Bessel functions to be able to do this simple problem!)
   - What is the total cross section in the extreme low-energy limit $k \to 0$.

3. (a) Express the condition of non-hermiticity of $\hat{H}$ as a property of the matrix elements of $\hat{H}$.

   (b) What is the action of parity operator ($\hat{P}$) and time reversal operator $\hat{\mathcal{T}}$ on position ($\hat{x}$) and momentum ($\hat{p}$) operator?

   (c) Show that for $\hat{\mathcal{T}}$ to preserve the canonical commutation relation between $\hat{x}$ and $\hat{p}$, $\hat{\mathcal{T}}$ has to be anti-linear.

   (d) What are the properties of eigenvalues of a $\hat{P}\hat{T}$ symmetric Hamiltonian?

   (e) Is $\hat{H} = \hat{p}^2 + i(\hat{x})^4$ a $\hat{P}\hat{T}$ symmetric?

4. For a two-level system, let the $H = \begin{pmatrix} \alpha + i\beta & 0 \\ 0 & \alpha - i\beta \end{pmatrix}$. Similar to what you have learnt in the class, let the action of $\hat{\mathcal{T}}$ on $\hat{H}$ be given by

   $\hat{\mathcal{T}}\hat{H}\hat{\mathcal{T}}^{-1} = \begin{pmatrix} \alpha - i\beta & 0 \\ 0 & \alpha + i\beta \end{pmatrix}$.

   (a) What is the action of $\hat{\mathcal{T}}$ on the basis states $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

   (b) What is the action of the identity operator $\mathbb{I}$ on the basis states $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

   (c) From (a) and (b), can we conclude that $\hat{T}$ is equal $\mathbb{I}$. Give reason to your answer.